CDS526: Artificial Intelligence Based Optimization

Lecture 9: Multi-objective Evolutionary Optimisation (Part II)

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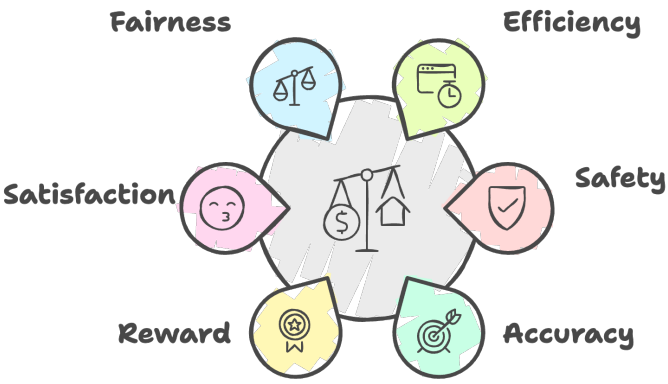
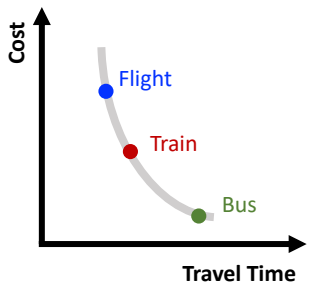
Objectives to Optimize

- Sometimes, one single objective
- Sometimes, multiple (competing) objectives





2

Multi-objective Optimisation Problem (MOP)

- Compared to "optimisation" that we have seen previously:
  - More than one (competing) objective to be optimised.
  - with or without constrains.

$$\begin{aligned} \min/\max F(\boldsymbol{x}) &= \{f_1(\boldsymbol{x}), \dots, f_m(\boldsymbol{x})\} && \text{<- } m \text{ objectives} \\ \textit{subject to} \quad g_j(\boldsymbol{x}) &\geq 0, \qquad j = 1,2,\dots,J && \text{<- Inequality constraint(s)} \\ h_k(\boldsymbol{x}) &= 0, \qquad k = 1,2,\dots,K && \text{<- Equality constraint(s)} \\ x_i^{(L)} \leq x_i &\leq x_i^{(U)}, \qquad i = 1,2,\dots,I && \text{<- Bounds, if any} \end{aligned}$$

where

- $\boldsymbol{x}$ is a vector of *continuous*, *discrete* or *mixed* variables,
- $x_i^{(L)}$ and $x_i^{(U)}$ are lower and upper bound of the i-th coordinate of $\boldsymbol{x}$.

Example

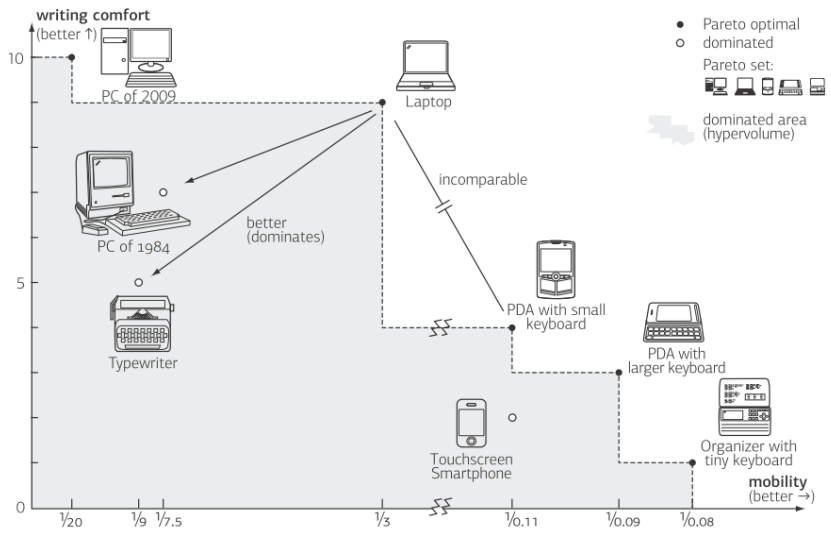
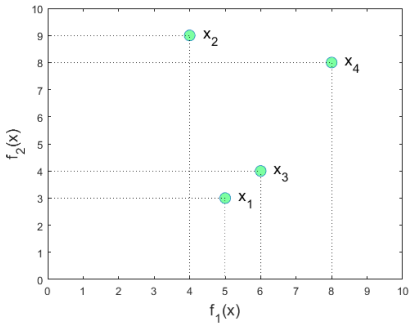


Figure from the presentation "*Two_Arch2: An Improved Two-Archive Algorithm for Many-Objective Optimization*" by Handing Wang

Pareto Dominance (帕累托支配)

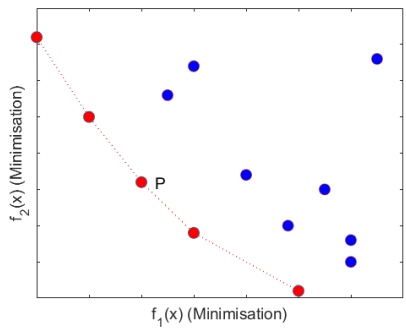
- **x_a dominates x_b** if
  - Solution x_a is no worse than x_b in all objectives
  - And solution x_a is strictly better than x_b in at least one objective.
  - Denoted as $x_a \preceq x_b$ if minimisation.
- **x_a dominates x_b $\Leftrightarrow$ x_b is dominated by x_a**

- Example: $\min F(x) = (f_1(x), f_2(x))$
  - x_1 dominates x_3? **YES**
  - x_1 dominates x_4? **YES**
  - x_2 dominates x_4? **NO**
  - x_2 dominates x_1? **NO**



Pareto Front (帕累托前沿)

- **Non-dominated solution set**: *among a set of solutions P*, the set of solutions that are not dominated by any member of P.
- **Pareto optimal set**: the non-dominated set of *the entire feasible decision space*.
- **Pareto optimal front**: the boundary defined by the set of all points mapped from the Pareto optimal set.

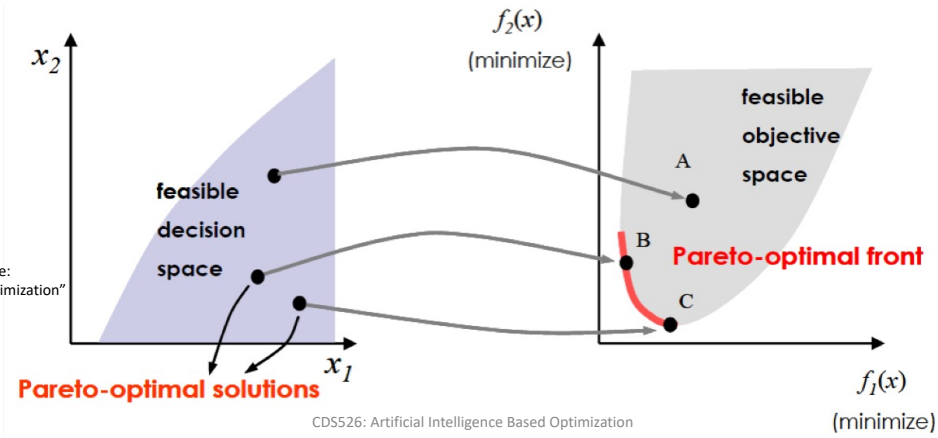


Non-dominated solutions: red points
Pareto optimal front: dashed red curve
Blue points are dominated by red points

Pareto Optimal Solutions

- Pareto optimal **set** is in the decision space (决策空间).
- Pareto optimal **front** is in the objective space (目标空间).

Image source:
"Multi-Objective Optimization"
by K. Deb

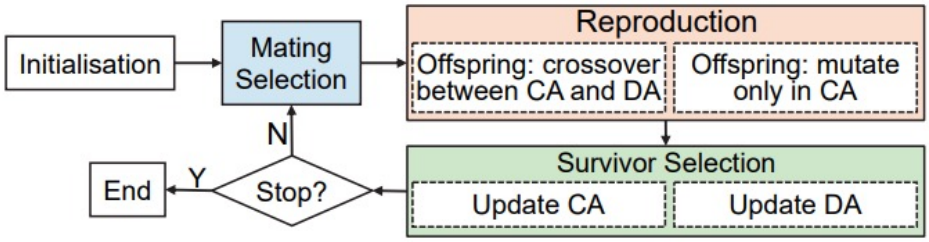


Outline

- **An Interactive Demo of Two_Arch2**
- MOEA/D

Improved Two-Archive Algorithm: Two_Arch2

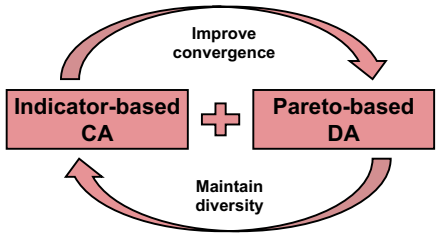
- Two-Archive algorithm2 (**Two_Arch2**) has two archives, convergence archive (CA) and divergence archive (DA), to promote convergence and diversity separately.



Handing Wang, Licheng Jiao, and Xin Yao. "Two Arch2: An improved two-archive algorithm for many-objective optimization". In: *IEEE Transactions on Evolutionary Computation* 19.4 (2015), pp. 524–541
Zhenshou Song et al. "A Kriging-Assisted Two-Archive Evolutionary Algorithm for Expensive Many-Objective Optimization". In: *IEEE Transactions on Evolutionary Computation* 25.6 (2021), pp. 1013–1027. DOI: 10.1109/TEVC.2021.3073648

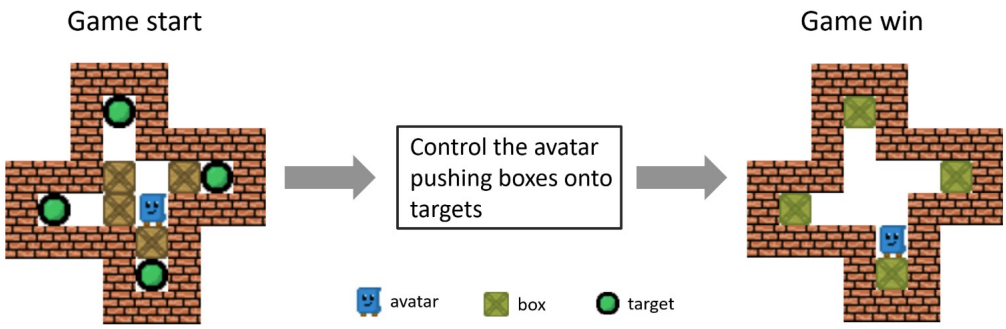
Two_Arch2: Main Steps

1. Initialisation.
2. Output DA if the stopping criterion is met, otherwise continue.
3. Generate new solutions from CA and DA by crossover and mutation.
4. Update CA and DA separately, go to 2.



Main ideas:
- The quality indicator $I_{\varepsilon+}$ can encourage convergence.
- The Pareto dominance can promote diversity.

Search for Sokoban Levels with Two_Arch2



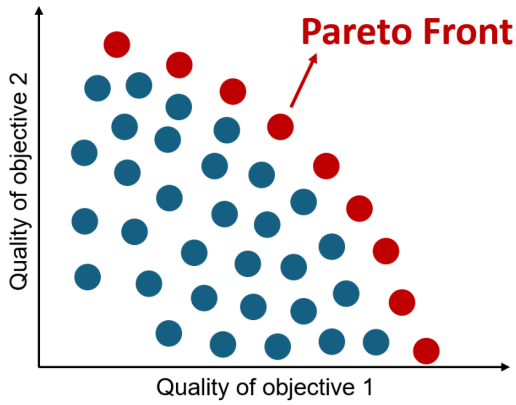
Zhang, Qingquan, Yuchen Li, Yuhang Lin, Handing Wang, and Jialin Liu. "Interpreting Multi-objective Evolutionary Algorithms via Sokoban Level Generation." In 2024 IEEE Conference on Games (CoG), pp. 1-2. IEEE, 2024.
https://aingames.cn/demo/mopcg/

Two Objectives in The Demo

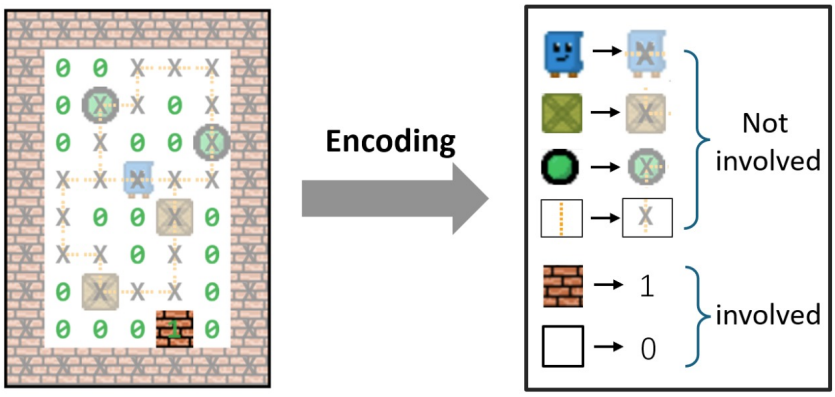
$$max\ \{f_1(level), f_2(level)\}$$

where

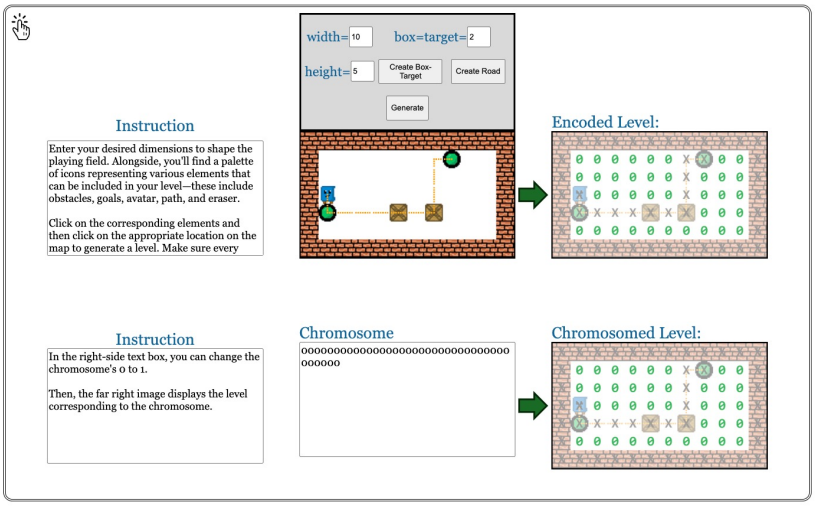
- $f_1(level)$ measures the proportion of empty tiles in a $level$.
- $f_2(level)$ assesses spatial diversity with values from 0 to 1, representing how different each row in a $level$ is.



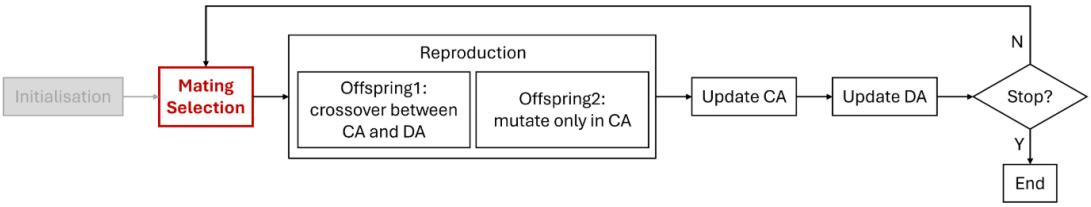
Representation of *Chromosome*



Level Setups and its Representation



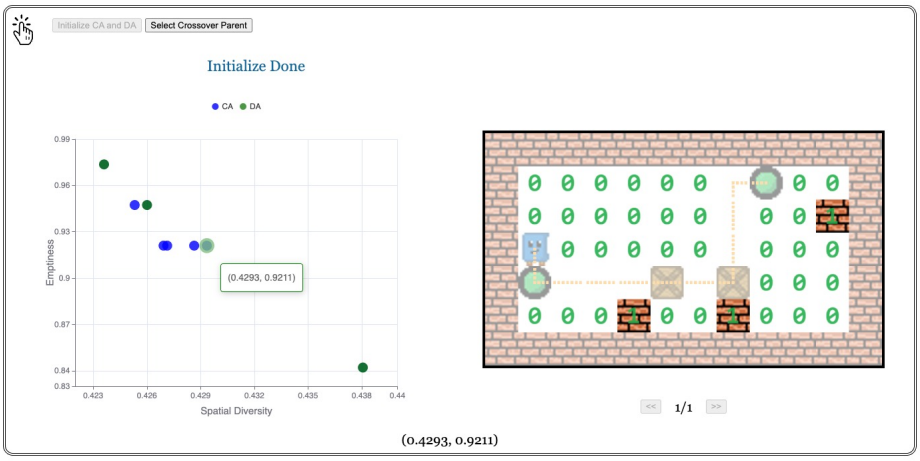
Mating Selection



Crossover Parent Selection

Divergence Archive

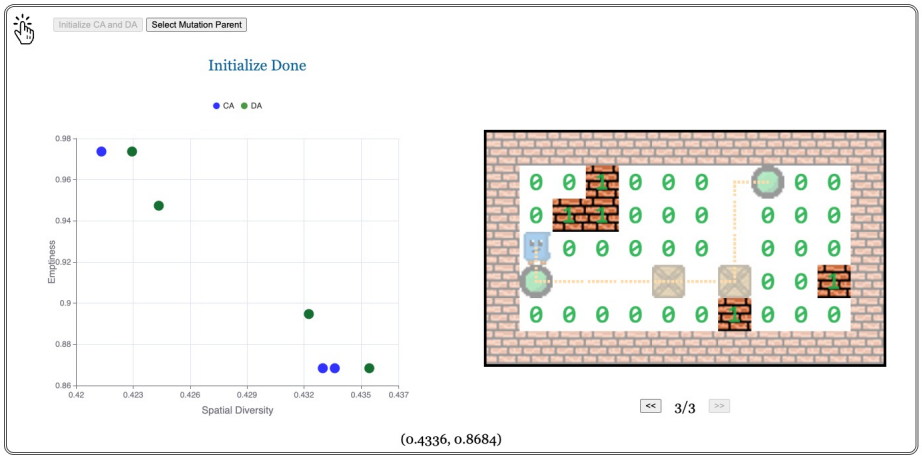
Convergence Archive



Mutation Parent Selection

Divergence Archive

Convergence Archive



(0.4336, 0.8684)

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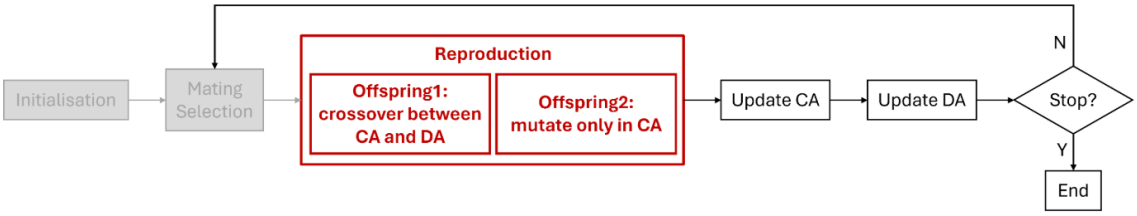
Reproduction



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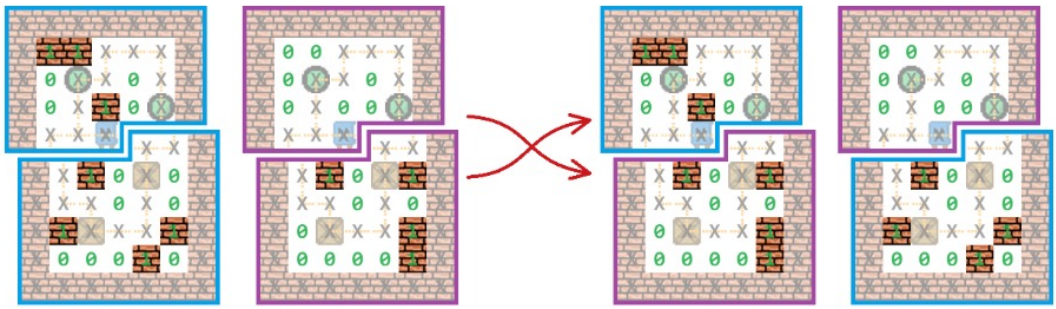
Crossover



Randomly choose a location
Divide level into half

Exchange half with each other

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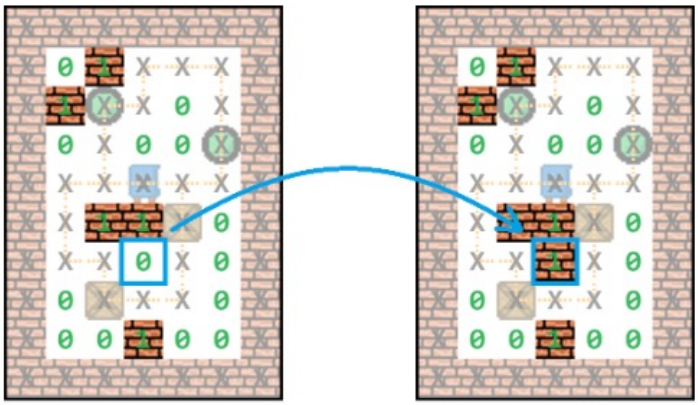
Mutation



Randomly choose a gene

wall → floor
floor → wall

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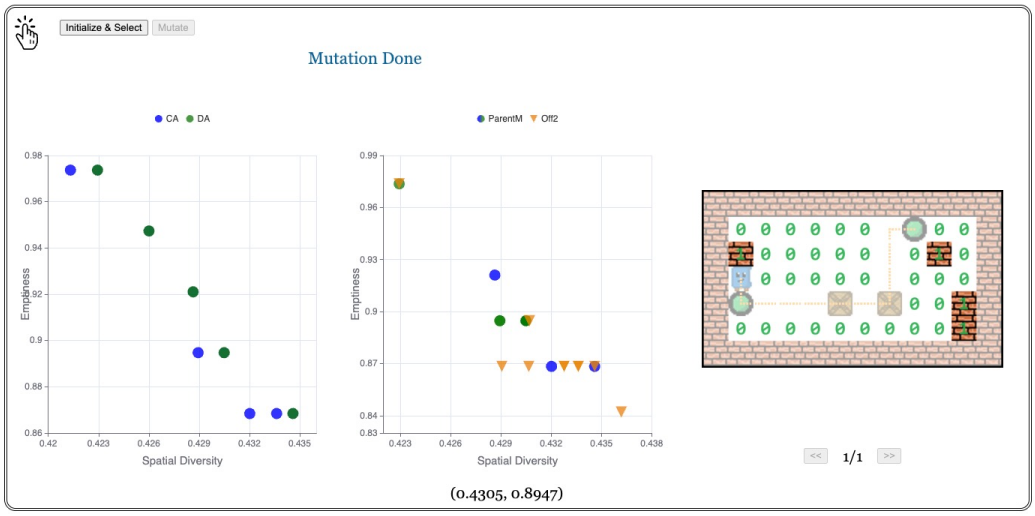
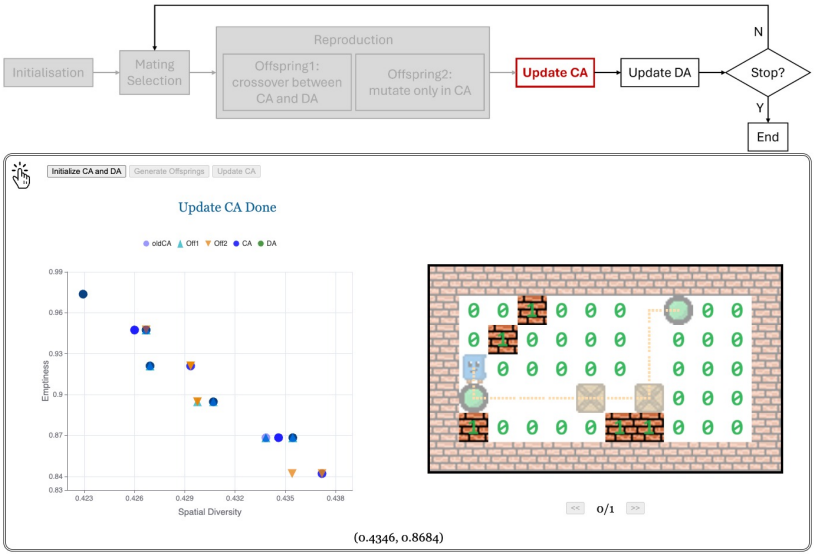
Mutation



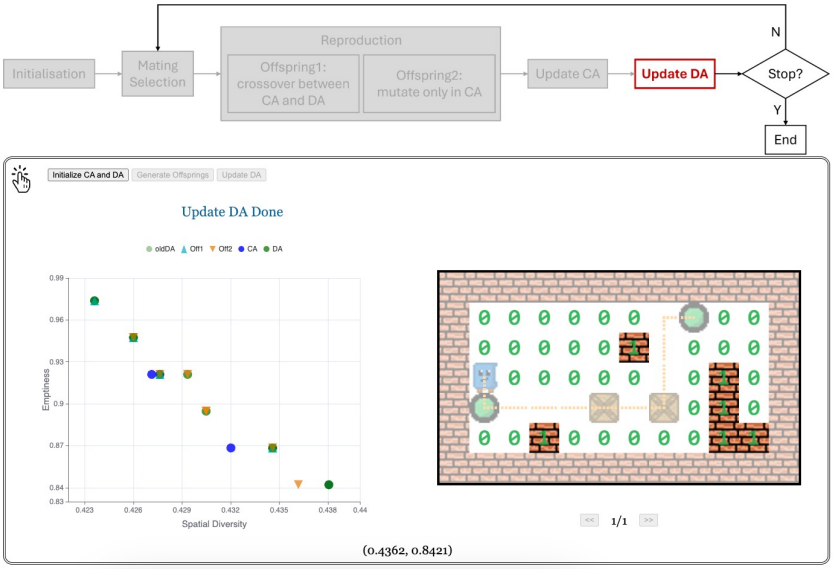
Update CA and DA

- The non-dominated offspring that dominate any solution in either CA or DA (the non-dominated solution with domination) are added to CA.
- The non-dominated offspring that dominate no solution in both CA and DA (the non-dominated solution without domination) are added to DA.
- Two_Arch removes extra solutions from DA according to their distances to CA.

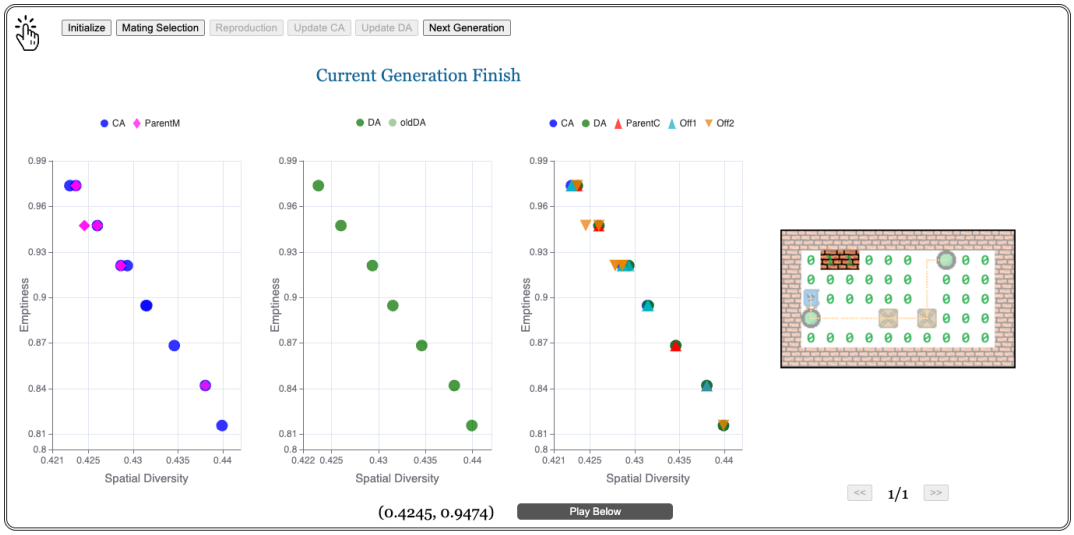
Update Convergence Archive



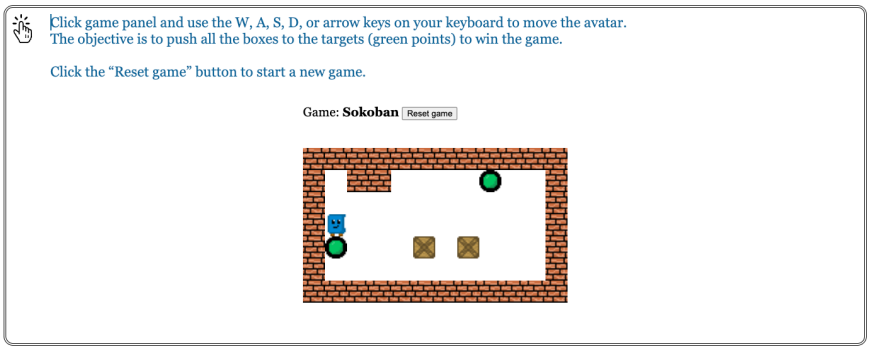
Update Divergence Archive



Overall Process



Play Generated Levels



Two_Arch2: An improved **two-archive** algorithm for many-objective optimization

H Wang, L Jiao, X Yao - IEEE transactions on evolutionary computation, 2014

☆ Cite Cited by 617 Related articles All 9 versions

Zhang, Qingquan, Yuchen Li, Yuhang Lin, Handing Wang, and Jialin Liu. "Interpreting Multi-objective Evolutionary Algorithms via Sokoban Level Generation." In 2024 IEEE Conference on Games (CoG), pp. 1-2. IEEE, 2024.

https://aingames.cn/demo/mopcg/

Outline

- **An Interactive Demo of Two_Arch2**
- **MOEA/D**

Multi-objective Optimisation Evolutionary Algorithms (MOEAs)

- Major categories of MOEAs:
  - **Pareto dominance** based: NSGA-II, SPEA2, PAES2, ...
  - **Performance indicator** based: SMS-EMOA, HyPE, ...
  - **Decomposition** based: MOEA/D, …
    - Decompose the task of approximating the Pareto Front into several subtasks. Each subproblem can be single-objective or multi-objective.
- Some algorithms may fall into more than one category.
  - Example: Two_Arch2 is indicator and Pareto-based

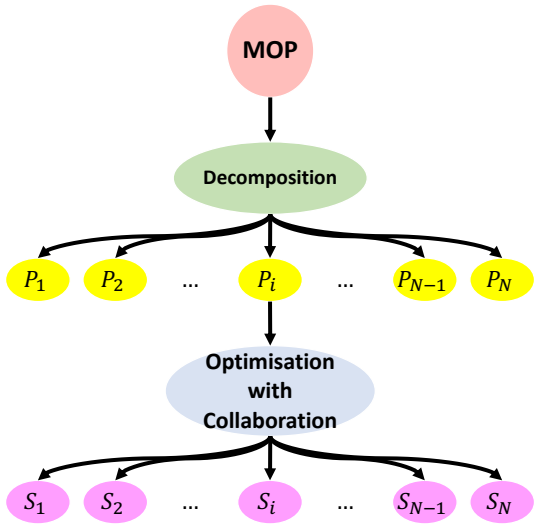
Decomposition Based Evolutionary Multi-objective Optimisation: MOEA/D

- **MOEA/D=Decomposition + Collaboration**
  - **Decomposition** (from traditional optimisation)
    - Decompose the task of approximating the Pareto Front into N subproblems. Each subproblem can be single-objective or multi-objective.
  - **Collaboration** (from swarm Intelligence and evolutionary computation)
    - N agents (procedures) are used. Each agent is for a different subproblem.
    - These N subproblems are related to one another. N agents can solve these subproblems in a collaborative manner.

Materials from "MOEA/D: Multi-Objective Evolutionary Algorithm Based on Decomposition, WCCI (CEC) 2024 Tutorial, Qingfu Zhang (June 30, 2024)" with author's permission

Major Issues

- *Problem decomposition* and *neighbourhood structure*.
- *Search method used by each agent*: memory and reproduction operators.
- *Collaboration*: mating selection and replacement.

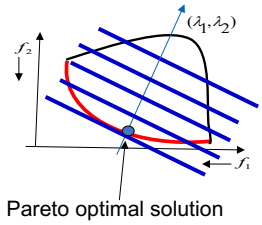


Problem Decomposition

- **Weighted sum approach:**

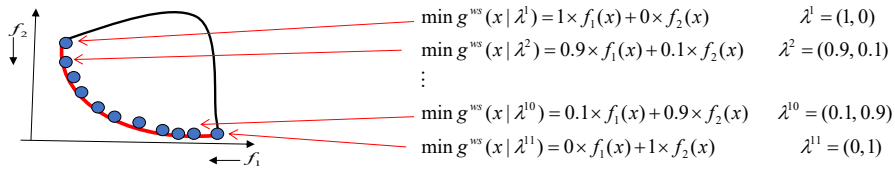
$$min\ g^{ws}(x|\lambda) = \lambda_1 f_1(x) + \lambda_2 f_2(x)$$

  - where $\lambda_1 + \lambda_2 = 1$ and $\lambda_1, \lambda_2 \geq 0$.
  - Works only for convex pareto-front.



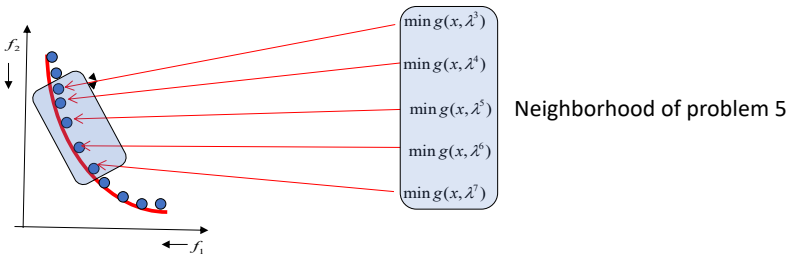
- **Approximation of the Pareto Front**

⇒ N single-objective optimization problems with N pairs of λ_1, λ_2.



Neighbourhood Structure

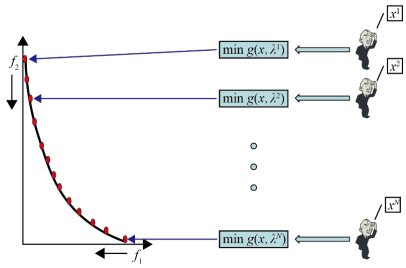
- ***Neighbourhood***: measure the relationship among subproblems
  - Two subproblems are neighbours if their weight vectors are close.
  - Neighbouring subproblems should have similar objective functions and thus similar optimal solutions with high probability.



- Many different ways for defining neighbourhood structure.

Search Algorithm Used by Each Agent

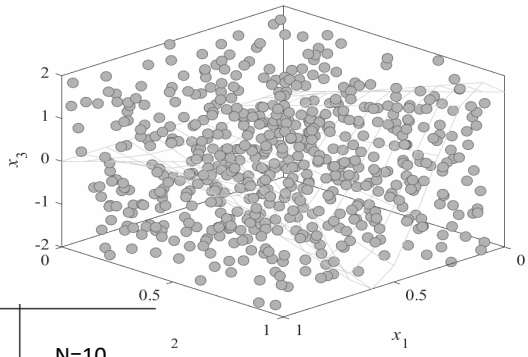
- At each generation, each agent does the following:
  1. **Mating selection**: obtain the current solutions of some neighbours (collaboration).
  2. **Reproduction**: generate a new solution by applying reproduction operators on its own solution and borrowed solutions.
  3. **Replacement**:
     1) replace its old solution by the new one if the new one is better than old one for its objective.
     2) pass the new solution on to some of its neighbours, each of them replaces its old solution by this new one if it is better for its objective. (collaboration, neighbourhood)



Decomposition, Neighborhood, Memory

Simulation Results

- Number of subproblems: 595
- Reproduction operators: DE+Mutation
- Number of generations: 500
- Test instance:



F6	$f_1 = \cos(0.5x_1\pi)\cos(0.5x_2\pi) + \frac{2}{\lvert J_1\rvert}\sum_{j\in J_1}(x_j - 2x_2\sin(2\pi x_1 + \frac{j\pi}{n}))^2$ $f_2 = \cos(0.5x_1\pi)\sin(0.5x_2\pi) + \frac{2}{\lvert J_2\rvert}\sum_{j\in J_2}(x_j - 2x_2\sin(2\pi x_1 + \frac{j\pi}{n}))^2$ $f_3 = \sin(0.5x_1\pi) + \frac{2}{\lvert J_3\rvert}\sum_{j\in J_3}(x_j - 2x_2\sin(2\pi x_1 + \frac{j\pi}{n}))^2$ where $J_1 = \{j \mid 3 \le j \le n$, and $j-1$ is a multiplication of $3\}$, $J_2 = \{j \mid 3 \le j \le n$, and $j-2$ is a multiplication of $3\}$, $J_3 = \{j \mid 3 \le j \le n$, and j is a multiplication of $3\}$ Its PS is $x_j = 2x_2\sin(2\pi x_1 + \frac{j\pi}{n}), j = 3, \ldots, n$.	N=10 $[0,1]^2 \times [-2,2]^{n-2}$

- Very hard and the optimal solution set is very complicated.

Li, H., & Zhang, Q. (2008). Multiobjective optimization problems with complicated Pareto sets, MOEA/D and NSGA-II. IEEE transactions on evolutionary computation, 13(2), 284-302.

Dynamical Resource Allocation

- Different subproblems (agents) require different amounts of computational resources.
- Each subproblem (agent) has a utility value, which measures the likelihood of further improvement:

$$\text{Utility} = \frac{\text{The amount of improvement obtained}}{\text{The amount of computational resouces used}}$$

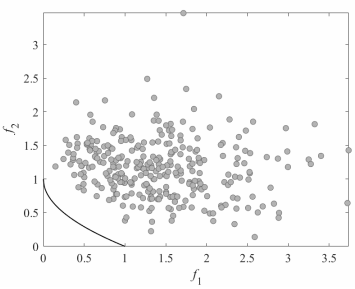
- At each generation, a small number of agents are selected based on utility values and receive computational resources.

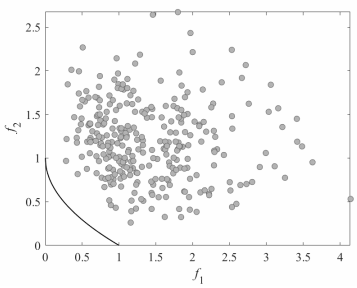
Simulation Results

- Number of subproblems: 300
- Reproduction operators: DE+Mutation
- Number of generation: 100
- Test instance:

F3	$f_1 = x_1 + \frac{2}{\lvert J_1 \rvert}\sum_{j\in J_1}(x_j - 0.8x_1\cos(6\pi x_1 + \frac{j\pi}{n}))^2$ $f_2 = 1 - \sqrt{x_1} + \frac{2}{\lvert J_2 \rvert}\sum_{j\in J_2}(x_j - 0.8x_1\sin(6\pi x_1 + \frac{j\pi}{n}))^2$ where J_1 and J_2 are the same as those of F1. Its PS is $x_j = \begin{cases} 0.8x_1\cos(6\pi x_1 + \frac{j\pi}{n}) & j \in J_1 \\ 0.8x_1\sin(6\pi x_1 + \frac{j\pi}{n}) & j \in J_2 \end{cases}$	n =10 $[0,1]\times[-1,1]^{n-1}$

MOEA/D





MOEA/D with dynamic resource allocation

Remarks

- Diversity among subproblems leads to diversity among $\{\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^N\}$.
- MOEA/D has a well-organized memory.
- Each agent can adopt other search methods such as heuristics (e.g., EDA, ACO and tabu search) or gradient based search.
- Agents can be divided into several groups. Each group share some common information such as probability models.
- There are many more decomposition methods and variants of MOEA/D!

MOEA/D Resources

- MOEA/D Web: https://sites.google.com/view/moead

MOEA/D: A multiobjective evolutionary algorithm based on decomposition
Q Zhang, H Li - IEEE Transactions on evolutionary computation, 2007
☆ 99 Cite Cited by 11336 Related articles All 23 versions

Summary

- MOEA/D= Decomposition + Collaboration, a methodology for multi-objective optimisation.
- Key design issues: decomposition, search method for each agent, collaboration.
- Different single optimisation methods can be readily used in MOEA/D framework.
- Multiobjective optimisation methods can also be integrated in MOEA/D framework.

Application Areas of MOO *

- **Engineering applications**
  - Electrical engineering [73]
  - Hydraulic engineering [63]
  - Structural engineering [37]
  - Aeronautical engineering [4]
  - Robotics and Control [70, 60]
- **Industrial applications**
  - Design and manufacture [47]
  - Scheduling [39]
  - Management [66]
- **Scientific applications**
  - Chemistry [57]
  - Physics [58]
  - Medicine [55]
  - Computer science [34]

* C. A. Coello Coello, "Evolutionary multi-objective optimization: a historical view of the field," in *IEEE Computational Intelligence Magazine*, vol. 1, no. 1, pp. 28-36, Feb. 2006, doi: 10.1109/MCI.2006.1597059.

All citation numbers listed on left refer to the references in the paper listed above.

Tools for MOO

- PlatEMO
- Pymoo: https://pymoo.org/
- Resources for MOEA/D: https://sites.google.com/view/moead/resources?authuser=0
- … …

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  - Presentation "*MOEA/D: Multi-Objective Evolutionary Algorithm Based on Decomposition*" by Prof. Qingfu Zhang with permission.

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